# Asmaa Shahine

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## Education

**Bachelor of Science in Aerospace Engineering** *2022 – June 2027*
Zewail City University of Science and Technology, Egypt

**Minor in Artificial Intelligence** *2025 – June 2027*
Zewail City University of Science and Technology, Egypt

## Graduation Project

**Autonomous Aerial Robotic System for NDT Inspection** *2026 – Present*
*Bachelor's Graduation Project – Aerospace Engineering*
Currently developing a multirotor UAV integrated with a robotic arm for autonomous non-destructive testing (NDT) inspection of industrial structures, focusing on aerial manipulation, flight stability, and inspection system integration.

## Professional Experience

**Egyptian Space Agency Internship** *Sep 2026 – Dec 2026*
*Egyptian Space Agency – On-site*
Internship in the aerospace and space technology field.

**Aircraft Factory Internship** *Aug 2026 – Sep 2026*
*Arab Organization for Industrialization (AOI)*
Industrial training at the Aircraft Factory.

**Engine Factory Internship** *Sep 2026 – Oct 2026*
*Arab Organization for Industrialization (AOI)*
Industrial training at the Engine Factory.

**University Lab – CNC Foam Laser Cutter for Airfoil Fabrication** *Jul – Oct 2025*
*Mechanical Design and Motion Control Project*
Designed and built a CNC foam laser cutter for airfoil fabrication,

**University Lab –UAV Design, Propulsion Selection, and Flight Testing** *Jul – Oct 2025*
Designed and developed a fixed-wing UAV from conceptual design to flight testing, conducting aerodynamic and stability analyses to select suitable propulsion and control components. Integrated and validated the UAV systems through successful flight trials.

**Advanced Aircraft Maintenance and Gas Turbine Engine Training** *Aug 2024*
Completed advanced training in aircraft maintenance and gas turbine engine systems.

**Gas Turbine Engine Training** *Aug 2024*
Completed technical training focused on gas turbine engines.

**Aircraft Maintenance Engineering Training** *Aug 2024*
Completed hands-on aircraft maintenance training covering maintenance procedures, troubleshooting, and repair practices.

## Project Experience

**ROTORTHON 2024 – VTOL Urban Air Mobility Design** *2024*
*Team Member, Aerospace Engineering Club* Achieved 4th place worldwide by designing a sustainable and efficient VTOL concept focused on urban air mobility, noise reduction, safety, and environmental impact.

**CFD Simulation for Gas Turbine Blade Cooling** *2025*
Conducted CFD-based thermal analysis of gas turbine blade cooling using ANSYS to evaluate temperature gradients and improve cooling efficiency.

**CFD Simulation of Flow in Curved Pipe** *2024*
Analyzed fluid flow in curved pipe geometries using ANSYS Fluent, evaluating velocity and pressure distributions to identify laminar and turbulent flow regions.

**HAWT Blade CFD and Structural Analysis** *2025*
Performed integrated CFD, structural stress, and modal analyses of a HAWT blade to evaluate aerodynamic performance, structural integrity, and vibration behavior, optimizing efficiency and safety.

**Joukowski Airfoil Simulation and CFD Validation** *2025*
Developed and validated a numerical Joukowski airfoil flow model using conformal mapping and ANSYS CFD, comparing inviscid and viscous results to evaluate pressure distributions and aerodynamic performance.

**Ramjet Engine Design and Performance Optimization** *2025*
Designed and optimized a supersonic ramjet engine through gas-dynamic analysis of shock waves, combustion, heat addition, and nozzle expansion to maximize thrust performance.

**Design and Prototyping of a 3-DOF Robotic Manipulator** *2025*
Designed, simulated, and fabricated a 3-DOF robotic manipulator, integrating kinematic modeling, SolidWorks CAD, 3D printing/CNC manufacturing, and microcontroller-based motor control.

**Smart Student Performance Prediction System (Machine Learning)** *2025*
Developed and evaluated machine learning models to predict student academic performance using a 20,000-sample dataset, applying data preprocessing, feature engineering, and predictive analytics.

**Active Suspension System Modeling and Control Design** *2025*
Modeled and analyzed a quarter-car active suspension system and designed state-feedback and LQR controllers using MATLAB/Simulink to enhance stability and ride comfort.

**Jet Engine Research and Cycle Analysis** *2024*
Researched various jet engine types and configurations and conducted thermodynamic cycle analyses of different propulsion systems.

**Advanced Shaft Design Project** *2025*
Designed and optimized a shaft system, including material selection, design calculations, and selection of suitable gears and bearings.

**Siphon System & Venturi Aerator Design** *2024*
Designed and modeled a siphon system and Venturi aerator using CAD software, validating performance through simulations and physical prototypes.

**Glider Design Project** *2023 – 2024*
Analyzed glider dimensions and performance using engineering software and fabricated balsa wood airfoil components to optimize aerodynamic performance.

**Python Game Development Projects** *2023*
Developed interactive Python games, including Sudoku, Tic-Tac-Toe, and shape assembly, featuring scoring systems, zoom functionality, and save features.

**FrozenLake Search-Based AI Agent** *2026*
Developed a FrozenLake AI agent using BFS, DFS, UCS, Greedy Search, and A* algorithms, comparing path optimality, cost, and execution efficiency.

**Deep Q-Network Reinforcement Learning Agent for CartPole** *2026*
Developed and trained a Deep Q-Network (DQN) reinforcement learning agent for CartPole control using Python, experience replay, and reward optimization to achieve stable decision-making.

**Flight Dynamics, Stability Analysis, and Autopilot Control Design** *2026*
Modeled the Lockheed JetStar/C-140 aircraft using NASA CR-2144 data and performed flight dynamics, stability, modal, and time-response analyses in MATLAB. Designed a pitch-angle feedback autopilot that reduced long-period settling time from 409.83 s to 88.50 s, achieving a 78.4

**Thin-Walled Wing Structural Design for Utility General Aviation Aircraft** *2026*
Designed a thin-walled wing structure for a utility-category aircraft, optimizing airfoil selection and wing geometry using XFLR5 and analytical calculations. Developed a CAD model with spars, stringers, ribs, and skin, and applied Prandtl's Lifting-Line Theory in MATLAB to analyze lift distribution and aerodynamic performance.

## Leadership & Extra-curricular Activities

**AIAA Chapter Initiative** *2024*
*Vice Chairperson & Founding Member, Zewail City*

**Hult Prize Competition** *2025*
*Startup Team Member*

**Zewail City Festival (EGYPES)** *2022*
*Volunteer Participant*

**Mathematics Online Tutor** *2024 – 2026*
*Independent Tutor*

**Aerospace Instructor – Space Point Company** *2025*
*Technical Instructor*

## Skills & Languages

**Technical Skills:** ANSYS, ANSYS Fluent, CFD, FEA, Python, C++, MATLAB, CAD Software, Simulation Tools, Engineering Design Software, Microsoft Office Suite
**Core Competencies:** Aerospace Systems Design, Aircraft Maintenance, CFD, Thermodynamics, Aerodynamics, Mechanical Design, Structural Analysis, Control Systems, Project Management, Technical Communication, Team Leadership
**Languages:** Arabic (Native), English (Proficient)